

CHAPTER # 2**ATOMIC STRUCTURE****"ATOM"****DEFINITION:**

In 1808, John Dalton gave a theory about the structure of atom. A/c to this theory:

"Matter was made from very small indivisible particles called atoms."

Today it is well established that,

"Atoms are complex organization of matter and energy."

CHARACTERISTICS OF ATOM:

- ✓ Atom takes part in chemical reactions independently.
- ✓ Atom can be divided into a number of sub-atomic particles.
- ✓ Fundamental particles of atom are electron, proton and neutron.

DISCOVERY OF FUNDAMENTAL PARTICLES:

Evidence for the presence of electron, proton, and neutrons in the atom is derived through many experiments such as:

- ✓ Faraday's experiment indicates the existence of electron.
- ✓ Crook's tube experiment shows the presence of electron and proton in the atoms.
- ✓ Radioactivity further confirms the presence of electron and proton.
- ✓ Chadwick's experiment shows the presence of neutrons.

SUBATOMIC PARTICLES AND THEIR CHARACTERISTICS:**Electron:**

The subatomic particle electron was discovered by J.J. Thomson in 1897 A.D, while studying on cathode rays experiment. He called the cathode rays beam as electrons. Thereafter, Millikan with his "Oil Drop Experiment" calculated the charge on an electron.

Protons:

Protons are positively charged subatomic particles which are located in the nuclei of all elements. They were discovered by Goldstein (1886 A.D), a German Physicist, by using perforated cathode in Crook's discharge tube.

Neutron:

The discovery of neutrons was made by James Chadwick (1932 A.D), in an artificial radioactivity experiment. He bombarded alpha particles from Polonium on Beryllium, a stable element and noted

that some highly penetrating neutral particles were produced. These particles were named as neutrons by James Chadwick (1932 A.D)

"FARADAY'S EXPERIMENT"

(PASSAGE OF ELECTRICITY THROUGH SOLUTION)

INTRODUCTION:

(CLUE ABOUT ELECTRONS)

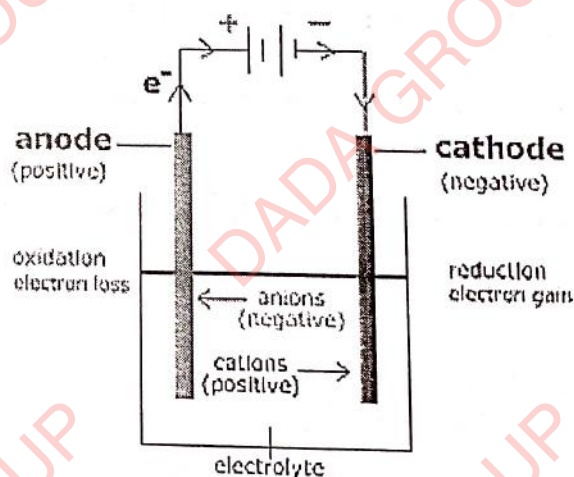
Earlier Davy had noticed that electric charges appear when solutions decompose by the passage of an electric current. Later this phenomenon was studied in greater details by Faraday.

ELECTROLYSIS:

"The process in which dissociated ions of an electrolyte move towards their respective electrodes on the passage of electricity and deposit their as neutral species is called electrolysis."

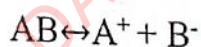
APPARATUS:

1. The process of electrolysis is done in an electrolytic cell containing electrolyte.
2. Two metallic plates are dipped in the electrolyte known as electrodes.
3. Both electrodes are connected to the battery.
4. The electrode which is connected to the positive terminal of battery is called anode and which is connected to the negative terminal is called cathode.

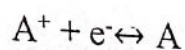


CELL REACTION:

On the passage of electricity following reaction takes place.

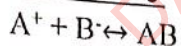
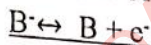
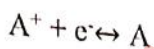
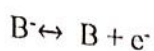


Reaction at cathode:



Reaction at anode:

Overall Reaction:



EXPERIMENTAL OBSERVATION:

When two metal plates called electrodes, are placed in an electrolytic solution and an electric current is passed, Faraday observed that,

- ✓ Electrolyte gave cations and anions on dissociate.
- ✓ Cations migrate towards cathode.
- ✓ Anion migrated toward anode.
- ✓ These ions neutralized their charges by gaining or losing electrons.
- ✓ Finally, they get deposited at the electrodes.

CONCLUSION:

The study of electrolysis gives a clue that atom may have electrical nature.

Hence there is some elementary unit of electric charges associated with these ions and it was found that its value is,

$$e^- = 1.60 \times 10^{-19} \text{ coulombs.}$$

The ions were observed to carry some integral multiply of this charge.

Later named by Stoney (1891), as "ELECTRON"

DISCOVERY OF ELECTRON

The first indication of the subatomic particles to be discovered was the electron. The knowledge about the electron was derived as a result of study of electric discharge in the discharged tube.

"DISCHARGED TUBE OR CROOKE'S TUBE EXPERIMENT"

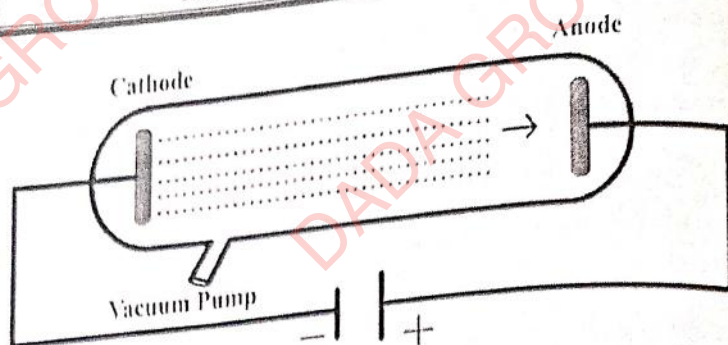
(PASSAGE OF ELECTRICITY THROUGH GASES AT LOW PRESSURE)

INTRODUCTION:

Julius Plucker first performed the discharged tube experiment. The nature and properties of the rays emitted in this tube was studied by William Crooke's and later on by J.J. Thomson.

STRUCTURE OF DISCHARGED TUBE:

The discharge tube is a long glass tube with circular electrodes sealed into each end. A small opening is made at its one side, which is connected to vacuum pump. The two electrodes are connected to the positive and negative terminals of a very high voltage battery.



WORKING / CONDUCTANCE OF ELECTRICITY IN THE DISCHARGED TUBE:

Gases do not conduct electricity under normal pressure. When a high voltage battery is applied across the electrodes, under reduced pressure then a series of changes occur in the discharge tube.

- ✓ **At 10mm Hg pressure:**
Spark of light passes like a flash.
- ✓ **At few mm Hg pressure:**
Metal electrodes glow and the space between them is dark called Crookes dark spaces.
- ✓ **At 1 mm Hg pressure:**
The tube is filled with a glow from positive electrode thus named positive column.
- ✓ **At 0.001 mm Hg pressure:**

The glow in the tube completely disappeared and the green fluorescence is observed in the tube at the anode side.

RESULT:

The color of light appears depend upon the nature of the gas filled in the discharge tube. In discharge tube it is obvious that some sort of radiation is passing between the two electrodes. This radiation consisted of vast swarm of particles emitted by the cathode and were named Cathode rays.

PROPERTIES OF CATHODE RAYS WITH EXPERIMENTAL PROOF:

Cathode rays are found to possess following properties.

✓ **PATH:**

Cathode rays travel in straight line.

EXPERIMENTAL PROOF:

When an object was placed in the path of cathode rays they produced sharp shadow of object at the opposite end.

✓ **CHARGE PARTICLES:**

They carry negative charge.

EXPERIMENTAL PROOF:

When cathode rays are produced between positive plate and negative plate then they deflected towards positive plate.

✓ MAGNETIC BEHAVIOUR:

These rays possess magnetic behavior.

EXPERIMENTAL PROOF:

When the magnetic field is applied onto the cathode rays they were deflected, showing magnetic behavior.

✓ MATERIAL PARTICLES:

Cathode rays are material particles and possess kinetic energy

EXPERIMENTAL PROOF:

Cathode rays caused the light paddle wheel to rotate when placed in their path.

✓ INDEPENDENCE:

The rays are independent of the nature of electrode, material or the gas enclosed in the tube.

EXPERIMENTAL PROOF:

The rays are not affected by the changing of electrode or gas in tube.

✓ CHARGE TO MASS RATIO:

Their charge to mass ratio resembles that of an electron.

EXPERIMENTAL PROOF:

The total charge of rays stream is divided by the total mass of it charge to mass ratio resembles to $1.602 \times 10^{-19} / 9.11 \times 10^{-28} = 1.76 \times 10^8 \text{ C/g}$

CONCLUSIONS:**✓ MATERIAL NATURE:**

Cathode rays are able to rotate a small pin wheel, able to cast sharp shadow of opaque body and produce fluorescence on zinc sulphide. Therefore these rays are considered to be material particles. The mass of these particles as calculated is 9.11×10^{-28} grams.

✓ CHARGE:

These rays are deflected towards the positive pole in the electromagnetic field. Therefore, these rays are considered to be negatively charged particles. The magnitude of charge on these particles is 1.60×10^{-19} coulombs.

✓ CHARGE TO MASS RATIO:

The charge to mass ratio of these particles was calculated by J.J. Thomson. Its value is similar to that of an electron. The e/m ratio is -1.76×10^8 coulombs/gram.

DISCOVERY OF PROTON
(DISCOVERY OF POSITIVE RAYS)**INTRODUCTION:**

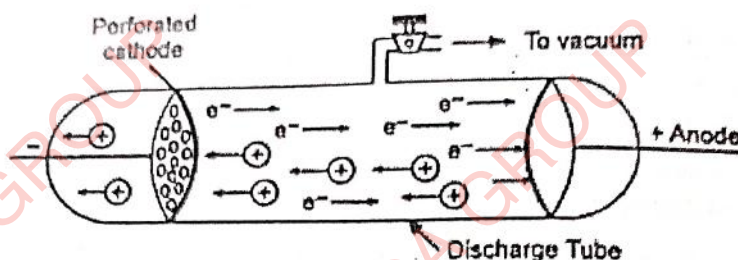
Experiments on cathode rays show that electrons are present in the atom. Since an atom is electrically neutral, therefore it must have some positively charged particle to balance the negative charge on electrons.

DETECTION OF POSITIVE RAYS (PROTONS):

During the discharge tube experiments, Gold Stein in 1886 observed that if a perforated cathode is used, then rays can pass through the perforations (canals) and accumulate behind the cathode. Since radiations are coming from anode, therefore, it was assumed that they must carry a positive charge. It was observed that these rays were originated in the middle of the discharged tube, when cathode rays collide with the gas molecules and ionize them.

RESULT:

Gold Stein named these rays as "POSITIVE RAYS". They are also known as "anode rays" and "canal rays".

**DEPEND ON NATURE OF GAS:**

The charge to mass ratio of these rays depends upon the nature of the gas in the discharge tube. The highest charge to mass ratio is obtained in case of hydrogen gas. Therefore hydrogen without electron (H^+) is the smallest positive particle of matter. This particle was given the name "PROTON" by GOLD STEIN.

PROPERTIES OF CANAL RAYS:

Canal rays are found to possess following properties.

✓ PATH:

They travel in the straight path, opposite to the cathode rays.

✓ MATERIAL PARTICLES:

They cause mechanical motion in a small pin wheel, placed in their path.

✓ CHARGE AND NATURE:

These rays are deflected by the electric as well as magnetic field. They bend towards negative pole in the electromagnetic field.

✓ CHARGE TO MASS RATIO:

The charge to mass ratio (e/m) of these rays depends upon the nature of gas used in the discharged tube. It is highest for hydrogen.

✓ MAGNITUDE:

The magnitude of charge of the canal rays was found to be 1.602×10^{-19} coulombs.

✓ MASS:

These particles have a mass 1836 times that of the electron are known as protons.

CONCLUSIONS:**✓ MATERIAL NATURE:**

Canal rays are able to rotate a small pin wheel, able to cast sharp shadow of opaque body produce fluorescence on zinc sulphide. Therefore these rays are considered to be material particles.

✓ **CHARGE:**

These rays are deflected towards the negative pole in the electromagnetic field. Therefore, these rays are considered to be positively charged particles. The magnitude of charge on these particles is 1.60×10^{-19} coulombs.

✓ **CHARGE TO MASS RATIO:**

The charge to mass ratio of these particles varies with the nature of gas used. Therefore it is concluded that these particles are produced from the atoms of gas in the discharged tube.

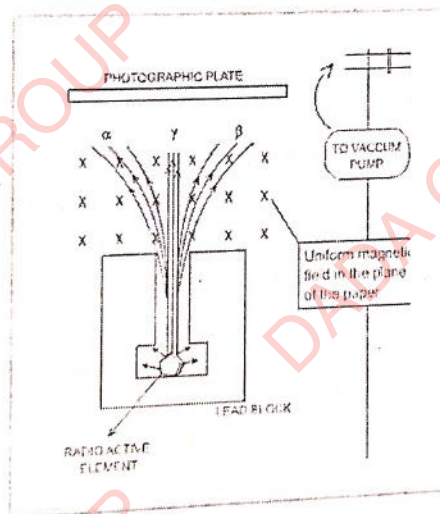
"RADIOACTIVITY"**INTRODUCTION:**

Professor Henri Becquerel, a French scientist was the first to discover radioactivity in 1896.

DEFINITION:

"The phenomenon in which certain heavy elements emit invisible radiations is called radioactivity."

The substances which give out these radiations are called radioactive substance.

DETECTION OF RADIOACTIVE RAYS:

Soon after the discovery of radium it was suspected that the rays given out by the radioactive substances were not all of one kind. Rutherford placed a piece of radioactive substance in a thick lead cup. The radiations were allowed to pass between to opposite poles of powerful electromagnet. He found that there were three distinct kinds of radioactive rays.

1. The rays bending towards the negative plate carries positive charge and were named as alpha (α) rays.
2. The rays bending towards the positive plate carries negative charge and were named as beta (β) rays.
3. The rays went un-deflected carries no charge and were named as gamma (γ) rays.

These rays were detected by placing photographic plate in their path. Each type of ray fogs the photographic plate at different places.

TYPES OF RADIOACTIVE RAYS:

There are three types of radioactive rays.

- ✓ Alpha (α) rays.
- ✓ Beta (β) rays.
- ✓ Gamma (γ) rays

"PROPERTIES OF RADIOACTIVE RAYS"

PROPERTIES OF ALPHA RAYS:

1. These rays consist of helium nuclei with double positive charged.
2. Their velocity range is 3×10^7 m/s to 3×10^6 m/s.
3. Being relative heavier in mass, they have less penetrating power. They can penetrate thin layer of solid such as mica or few centimeter of air. They are stopped by an aluminum foil that is 0.1 mm thick.
4. They are good ionizer of gases.
5. They blacken the photographic plate in dark.
6. They are deflected by electric and magnetic field.

PROPERTIES OF BETA RAYS:

1. These rays consist of negative charged particles.
2. Their e/m ratio is same as that of cathode rays, therefore they are named electrons.
3. Velocity of β -rays is from 9×10^7 m/sec to 27×10^7 m/sec.
4. Being relative lighter in mass, their penetrating power is far greater than alpha rays. They are stopped by an aluminum sheet that is 1 cm thick.
5. The beta particles have far less ionizing power than alpha rays.
6. They are deflected by electric and magnetic field but to the opposite direction of Alpha rays.

PROPERTIES OF GAMMA RAYS:

1. These rays are electromagnetic radiations with no charge.
2. They are similar to X-rays but smaller in wavelength.
3. γ - rays travel with the velocity of light that is 3×10^8 m/sec.
4. Their penetrating power is much greater than other two rays.
5. They are weak ionizer of gases.
6. These rays are emitted only after the parent nucleus has changed i.e. after the beta particles have been ejected.
7. They pass without deflection in electric and magnetic field.

"ARTIFICIAL RADIOACTIVITY"**DEFINITION:**

"When a stable element is bombarded with alpha particles or protons its nucleus becomes unstable and emits radiations to stabilize it. This induced radioactivity is called artificial radioactivity".

During the course of artificial radioactivity not only new elements are formed but also new subatomic particles like protons, neutrons are discovered.

"NATURAL AND ARTIFICIAL RADIOACTIVITY"

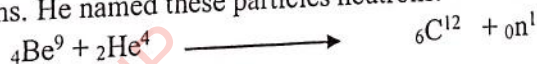
<u>NATURAL RADIOACTIVITY</u>	<u>ARTIFICIAL RADIOACTIVITY</u>
1. It is a spontaneous process. 2. Only the elements having atomic number Greater than 82 show this phenomenon. 3. New elements are produced after centuries. 4. The uses of natural radioactive elements are limited.	1. It has to be induced in a non-radioactive element. 2. The elements having atomic number lower than 82 are rendered to be radioactive. 3. New elements are produced in short times. 4. The uses of natural radioactive elements are very vast.

"DISCOVERY OF NEUTRON"**INTRODUCTION:**

In 1932 Chadwick discovered neutron by using the concept of artificial radioactivity.

DETECTION OF NEUTRON:

Chadwick bombarded the nucleus of beryllium with fast moving alpha particle and observed that radiations having high penetrating power were produced. These rays are neutral particles having mass equal to that of protons. He named these particles neutrons.



The above reaction is called (α , n) reaction.

RESULT:

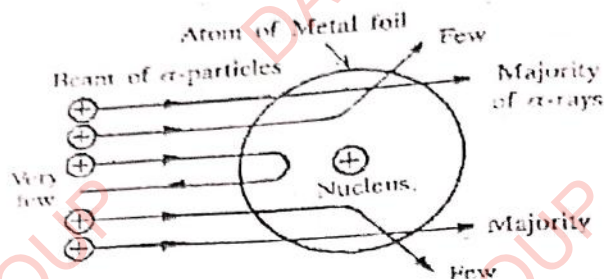
With the help of Artificial radioactivity another fundamental particle "NEUTRON" was discovered.

"RUTHERFORD'S ATOMIC MODEL"
(DISCOVERY OF NUCLEUS)**INTRODUCTION:**

Having known that atom contains electrons and protons. Rutherford proceeded to perform experiment to know how and where these particles are located in atom.

EXPERIMENT:

Rutherford bombarded a thin foil of metal (Gold) by streams of positively charge particles that is alpha (α) particles emitted from a radioactive source (polonium). A fluorescent screen is placed beyond the gold foil in order to check the path of alpha particles.

**OBSERVATION:**

It was found that:

1. Most of the alpha particles penetrated the foil and emerged undeflected on the other side.
2. About 1% of alpha particles show slight deviation in their paths
3. One out of 8000 suffered a deflection greater than 90° and emerged on the same side.

CONCLUSION:

From the above experiments, Rutherford reached to the following conclusions.

1. Since most of the α - particles passed un-deflected through the gold foil, therefore most of space of an atom is empty.
2. The deflection of few particles showed that an atom has a positively charged body at the Centre. He named it nucleus.
3. The nucleus of an atom is heavy; therefore whole mass of the atom is concentrated in the nucleus.
4. All the mass of an atom is concentrated at its center called nucleus.
5. As the atom is electrically neutral, therefore the electrons must be present in the outer nuclear region.
6. The electrons are continuously revolving around the nucleus so as to make them stabilize and prevent them from falling into the nucleus.

DEFECTS OF RUTHERFORD'S ATOMIC MODEL:

According to electromagnetic theory of radiation, any charged particle loses energy during its state of motion.

Now in Rutherford's atomic model;

1. **LINE SPECTRUM:**

Since electrons are revolving around the nucleus, they should lose energy continuously and a continuous spectrum should be obtained. But in actual case a line spectrum is observed.

2. **REVOLUTION OF ELECTRON:**

The revolving electron should come closer to the nucleus continuously and it should fall into it. But in case it never happens.

"PLANK'S QUANTUM THEORY"

(QUANTIZATION OF ENERGY)

INTRODUCTION:

This theory was proposed in 1900 by the German physicist MAX PLANK to account for the observed radiations from heated bodies.

POSTULATES:

Following are the main postulates of plank's quantum theory.

- ✓ The energy could not be absorbed or emitted by the atoms in any arbitrary quantity, but only in specified amount called QUANTA.
- ✓ An atom cannot change its energy continuously but only by a series of steps.
- ✓ The amount of energy depends upon the frequency (ν) of radiation absorbed or emitted. It is given by relation:

$$E = h \nu$$

Where, h = Plank's constant (6.625×10^{-34} J.S = 6.625×10^{-27} erg. Sec)

CONSEQUENCE:

The amount of energy gained or lost is quantized i.e., energy change occurs in small packets or multiple of those packets, $h \nu$, $2h \nu$, $3h \nu$...

"BOHR'S ATOMIC THEORY"

INTRODUCTION:

In order to remove the difficulties which arise in explaining the structure of the atom in the light of Rutherford's atomic theory, Neil Bohr developed a model and applied quantum theory to explain the behavior of electrons.

POSTULATES:

The main postulates of this theory are:

✓ **STATIONARY STATES:**

Electrons in an atom are revolving round the nucleus in certain circular orbits of fixed energy called energy levels or stationary states.

✓ **ENERGY OF ELECTRON:**

The energy of the electron in the orbits remains constant i.e. it neither emits nor absorbs energy.

✓ **EMMISSION OR ABSORPTION OF ENERGY:**

Electron emits or absorbs energy only in fixed amount or quanta. The energy change (ΔE) when the electron jumps from one orbit to another is given by;

$$\Delta E = E_2 - E_1$$

Where E_1 and E_2 are the energies of electron in the first and second energy states

✓ **ANGULAR MOMENTUM OF ELECTRON:**

The angular momentum of the revolving electron in any orbit is an integral multiple of $\frac{h}{2\pi}$ that is angular momentum is quantized.

$$mvr = n \frac{h}{2\pi}$$

Where, h = Plank's constant (6.625×10^{-34} J.S = 6.625×10^{-27} erg. Sec).

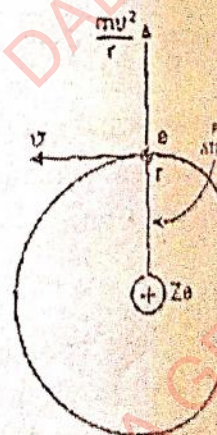
"APPLICATION OF BOHR'S THEORY"**IN HYDROGEN ATOM:**

By applying the concept of the quantum of energy and laws of classical mechanics Bohr worked a mathematical expression for the derivation of radius, energy, frequency and wave number of electron moving with a certain orbit in case of Hydrogen atom.

DETERMINATION OF RADIUS OF HYDROGEN ATOM:

Consider an electron of mass ' m ' and charge ' e ', revolving around the nucleus of charge ' Ze ' in orbit of radius ' r '.

The electrostatic force or centripetal force acting between the nucleus and the electron is given by:



According to coulomb's law:

$$F = \frac{k q_1 q_2}{r^2} \text{----- (i)}$$

Where,

q_1 = charge of nucleus = Ze

eq(i) becomes

$q_2 = \text{charge of electron} = e$
 $k = 1$ (in C.G.S. system)

$$F = \frac{(1)(Ze)(e)}{r^2}$$

$$F = \frac{Ze^2}{r^2} \text{ ----- (ii)}$$

Since,

Coulomb's force is actually the centripetal force which attracts the electron towards the nucleus.
 As a reaction equal and opposite centrifugal force acting on electron is given by;

$$F_c = \frac{mv^2}{r}$$

The two forces balance each other i.e.

$$F = F_c$$

$$\frac{Ze^2}{r^2} = \frac{mv^2}{r}$$

$$\frac{r}{r^2} = \frac{mv^2}{Ze^2}$$

$$r = \frac{Ze^2}{mv^2} \text{ ----- (iii)}$$

According to Bohr's theory;

$$mvr = \frac{nh}{2\pi}$$

$$v = \frac{nh}{2\pi} \times \frac{1}{mr}$$

$$v = \frac{nh}{2\pi mr} \text{ ----- (iv)}$$

Substituting eq(iv) in eq(iii)

eq(iii) becomes

$$r = \frac{Ze^2}{m\left(\frac{nh}{2\pi mr}\right)^2}$$

⇒

$$r = \frac{n^2 h^2}{4\pi^2 m Ze^2} \text{ ----- (v)}$$

This is the expression for radius of hydrogen orbit.

Hence the radius of 1st orbit of hydrogen atom is calculated as:

$$n = 1$$

$$h = 6.625 \times 10^{-34} \text{ J.sec}$$

$$m = 9.11 \times 10^{-31} \text{ kg}$$

$$Z = 1$$

$$e = 1.602 \times 10^{-19} \text{ C}$$

$\Rightarrow$ $\Rightarrow$

$$r = 0.529 \text{ \AA}$$

$$r = n^2 a_0$$

Where $n = 1, 2, 3, \dots$

$$1 \text{ \AA} = 10^{-10} \text{ m}$$

$$a_0 = 0.529 \text{ \AA}$$

DETERMINATION OF ENERGY:

We know that the total energy of electron in the first orbit is the sum of its kinetic energy and potential energy i.e.

$$E = K.E + P.E \text{----- (i)}$$

The K.E. of electron is given by,

$$K.E = \frac{1}{2} mv^2$$

Where,

m = mass of electron

V = velocity of electron

The potential energy of electron is given by;

$$P.E = \frac{-ze^2}{r}$$

z = atomic number

e = charge of electron

r = radius of orbit

Substituting these values in eq i we get;

$$E = \frac{1}{2}mv^2 + \frac{-Ze^2}{r} \text{----- (ii)}$$

From the derivation of radius we know that

$$\frac{Ze^2}{r^2} = \frac{mv^2}{r}$$

$$mv^2 = \frac{Ze^2}{r}$$

Multiply both sides by $\frac{1}{2}$ we get:

$$\frac{1}{2} mv^2 = \frac{Ze^2}{2r}$$

$\Rightarrow$ Eq(ii) becomes,

$$E = \frac{Ze^2}{2r} - \frac{Ze^2}{r}$$

$$E = \frac{-Ze^2}{2r}$$

We know that,

$$r = \frac{n^2 h^2}{4\pi^2 m Ze^2}$$

$$E = \frac{-ze^2}{2\left(\frac{n^2 h^2}{4\pi^2 mZe^2}\right)}$$

$$E = \frac{-2\pi^2 mZ^2 e^4}{n^2 h^2}$$

This is the expression for the energy of electron.

FREQUENCY:

Energy of first shell is given by;

$$E_1 = \frac{-2\pi^2 mZ^2 e^4}{n_1^2 h^2}$$

Energy of 2nd shell is given by;

$$E_2 = \frac{-2\pi^2 mZ^2 e^4}{n_2^2 h^2}$$

Change in energy is given by;

$$\Delta E = E_2 - E_1$$

$$\Delta E = \frac{-2\pi^2 mZ^2 e^4}{n_2^2 h^2} + \frac{2\pi^2 mZ^2 e^4}{n_1^2 h^2}$$

$$\Delta E = \frac{2\pi^2 mZ^2 e^4}{h^2} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

According to Bohr's theory,

$$\Delta E = h\nu$$

$$h\nu = \frac{2\pi^2 mZ^2 e^4}{h^2} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\nu = \frac{2\pi^2 mZ^2 e^4}{h^3} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{----- (iii)}$$

The above expression is used to find the frequency of light emitted during electron transition.

WAVE NUMBER:

According to wave theory

$$C = \nu\lambda$$

$$\nu = c/\lambda$$

eq(iii) becomes;

$$\frac{c}{\lambda} = \frac{2\pi^2 mZ^2 e^4}{h^3} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{1}{\lambda} = \frac{2\pi^2 mZ^2 e^4}{ch^3} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\bar{\nu} = \frac{2\pi^2 me^4 Z^2}{ch^3} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Since $\bar{\nu} = 1/\lambda$

In the above equation $\frac{2\pi^2 me^4}{ch^3}$ is constant, which is called Rydberg's constant and its value is,

$$R_H = 109678 \text{ cm}^{-1}$$

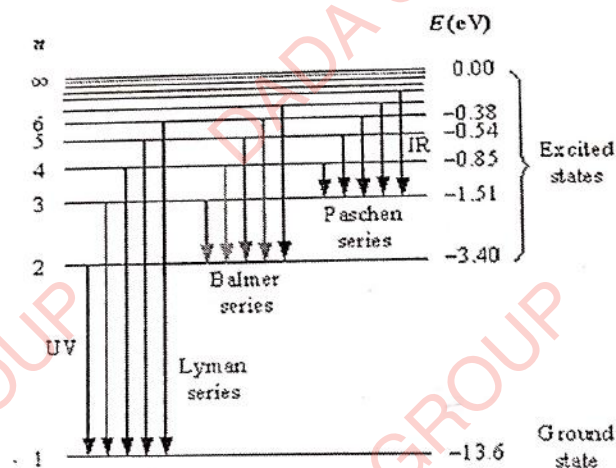
$$\bar{\nu} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

This is the expression for wave number of radiation.

"HYDROGEN SPECTRUM"

Historical background:

"The line spectrum obtained from excited electrons of Hydrogen atoms is called Hydrogen spectrum."



Lyman series:

- The radiations of this series were in ultraviolet region.
- This series is obtained when electron jumps from 2, 3, 4,.... To 1st orbit.
- Wave number $\nu = R_H [1/n_1^2 - 1/n_2^2]$ where $n_1=1$ $n_2=2, 3, 4, \dots$

Balmer series:

- The radiations of this series were in visible region
- This series is obtained when electron jumps from 3, 4, 5,.... To 2nd orbit.
- Wave number $\nu = R_H [1/n_1^2 - 1/n_2^2]$ where $n_1=2$ $n_2=3, 4, \dots$

Paschen series:

- The radiations of this series were in infrared region.
- This series is obtained when electron jumps from 4, 5, 6,.... To 3rd orbit.
- Wave number $\nu = R_H [1/n_1^2 - 1/n_2^2]$ where $n_1 = 3$ $n_2 = 4, 5, 6, \dots$

Brackett series:

- The radiations of this series were in far-infrared region.
- This series is obtained when electron jumps from 5, 6, 7, To 4th orbit.
- Wave number $\nu = R_H [1/n_1^2 - 1/n_2^2]$ where $n_1=4$ $n_2=5, 6, 7, \dots$

Pfund series:

- The radiations of this series were also in far-infrared region.
- This series is obtained when electron jumps from 6, 7, 8, To 5th orbit.
- Wave number $\nu = R_H [1/n_1^2 - 1/n_2^2]$ where $n_1=5$ $n_2=6, 7, 8, \dots$

"HEISENBERG'S UNCERTAINTY PRINCIPLE"**INTRODUCTION:**

In 1925, Heisenberg proposed a principle in order to accurate Bohr's explanation about the orbits.

PRINCIPLE:

"It is not possible to determine simultaneously the position (x) and momentum (p) of moving electron; if one is known exactly it would be impossible to know other exactly".

MATHEMATICALLY:

The product of their uncertainty is of the order of Planck's constant.

$$\Delta p_x \Delta x \approx h$$

Where,

Δp = uncertainty in momentum

Δx = uncertainty in position

Thus if one of the two i.e. Δp or Δx was known exactly, then the uncertainty in its determination could be zero and that of the other would become infinite.

EXPLANATION:

According to Bohr's theory, electron is a particle whose momentum and path is known precisely. But according to De Broglie, electron is regarded as wave. Therefore electron has dual nature when electron is suggested as wave it is not possible to measure simultaneously its exact position and momentum, at particular line.

CAN THIS BE APPLICABLE TO ALL OBJECTS?

The massive bodies seen in our daily life do not change their position and when they interact with electromagnetic radiations, hence for heavy bodies this principle has no significance, i.e., for big objects velocity and position can be measured accurately.

CONCLUSION DRAWN FROM HEISENBERG'S UNCERTAINTY PRINCIPLE OR DEFECT OF BOHR'S ATOMIC MODEL:

The following points are concluding from Heisenberg's principle;

1. An electron cannot be at rest.
2. The energy of an electron cannot be zero.
3. Particular location of the electron cannot be determined.

Thus, Bohr's orbit lose their meaning as circular paths, Heisenberg's principle explains the basic incompleteness of Bohr's atomic model.

"QUANTUM NUMBERS"

The various electrons are revolving around the nucleus of an atom have different positions and energies.

Therefore, complete 'address' of an electron is provided by a set of four index numbers, known as quantum numbers.

DEFINITION:

"A set of four index numbers, which are required to completely specify the position and energy of an electron, in an atom, is called quantum numbers."

The four quantum numbers and their significance have been described below.

a) PRINCIPAL QUANTUM NUMBER:

The principal quantum number is designated by n and n can have any positive integral number 1, 2, 3, ----.

SIGNIFICANCE:

- (i) It determines the energy of an electron. For example, when $n=1$, the electron has the lowest energy.
- (ii) It describes the average distance of an electron from the nucleus of an atom. For example, when $n=1$, the electron shell is close to the nucleus. In other words, the larger the value of n , the bigger the electron cloud.

b) AZIMUTHAL QUANTUM NUMBER:

This quantum number is designated by l which can have all integral values from 0 to $n-1$.
 $l = 0, 1, 2, \dots (n-1)$

For example,

- | | | |
|----------|---------------------|---------------------|
| If $n=1$ | then $l=0$ | ---- (one value) |
| If $n=2$ | then $l=0, 1$ | ---- (two values) |
| If $n=3$ | then $l=0, 1, 2$ | ---- (three values) |
| If $n=4$ | then $l=0, 1, 2, 3$ | ---- (four values) |

It is clear that the total values of l are equal to the number of principal quantum number.

SIGNIFICANCE:

- (i) It denotes the sub-shell sub-energy level in a given principal shell to which electron belongs.
- (ii) It describes the shapes of sub-energy levels. Thus when an electron has
- $l = 0$, it means it is said to be in 's-sub-shell'
 - $l = 1$, it means it is said to be in 'p-sub-shell'
 - $l = 2$, it means it is said to be in 'd-sub-shell'
 - $l = 3$, it means it is said to be in 'f-sub-shell'

c) MAGNETIC QUANTUM NUMBER:

This quantum number is designated by m and it can have the values from ' $-l$ through 0 to $+l$ '.
That is $m = -l, \dots, 0, \dots, +l$.

For example

- For s-sub-shell, $l=0$, m has one value, i.e., 0 . Hence s-sub-shell contains one orbital called s-orbital.
- For p-sub-shell, $l=1$, m has three values, i.e., $-1, 0, +1$. Hence p-sub-shell contains three orbitals called p-orbitals.
- For d-sub-shell, $l=2$, m has five values, i.e., $-2, -1, 0, +1, +2$. Hence d-sub-shell contains five orbitals called d-orbitals.
- For f-sub-shell, $l=3$, m has seven values, i.e., $-3, -2, -1, 0, +1, +2, +3$. Hence f-sub-shell contains seven orbitals called f-orbitals.

SIGNIFICANCE:

It determines preferred orientation of orbital in space.

d) SPIN QUANTUM NUMBER:

This quantum number is designated by s and it can have the values ' $+\frac{1}{2}$ or $-\frac{1}{2}$ '.

SIGNIFICANCE:

It determines the spin of orientation of the electron.

"ELECTRONIC CONFIGURATION"

DEFINITION:

"The distribution of electrons in various sub-energy levels and orbital is called electronic configuration".

RULES:

Different rules and principles govern this distribution.

- ✓ Pauli exclusive principle.
- ✓ Aufbau principle.
- ✓ $(n + l)$ rule.
- ✓ Hund's rule.

AUFBAU PRINCIPLE:

"The electrons enter in the sub-shell with the lowest energy."
OR

"The filling of electrons in the sub-energy levels of an atom takes place in the increasing order of energy starting with 1s."

The sequence of increasing orbital energy is:
1s, 2s, 2p, 3s, 3p, 4s, 3d, -----

(n + l) RULE:

"In building up the electronic configuration of the elements, the orbital with lowest value of (n + l) fills first; when two orbitals have the same value of (n + l) the orbital with the lower value of 'n' fills first."

PAULI'S EXCLUSION PRINCIPLE:

"No two electrons in an atom can have four sets of same quantum numbers."

HUND'S RULE OF MAXIMUM MULTIPLICITY:

When the degenerate orbitals are available, then

- Electron will enter in different orbitals with parallel spin.*
- If orbitals are partially filled then the coming electron will form pair with their electron with opposite spin.*

"SPECTRUM"**DEFINITION:**

"The band of colors into which the incident light breaks up is called the spectrum."

"EMISSION SPECTRUM"**DEFINITION:**

"When an element emits radiant energy, on passing this radiation through a prism it is separated into component wave lengths to form an image called emission spectrum."

Emission spectra are of two types:

- Continuous spectrum
- Line spectrum

"CONTINUOUS SPECTRUM"**DEFINITION:**

"Such a spectrum in which there is no line of demarcation between different colors is called continuous spectrum."

EXPLANATION:

When white light from the sun or any incandescent lamp is allowed to pass through a prism, the light is not only deviated but also breaks up into its constituent colors and gives a continuous spectrum on one end the least deviated is red and at other end is violet which suffers the maximum deviation.

"LINE SPECTRUM"**DEFINITION:**

"The spectrum consist of bright lines separated from each other by dark spaces is called line spectrum."

EXPLANATION:

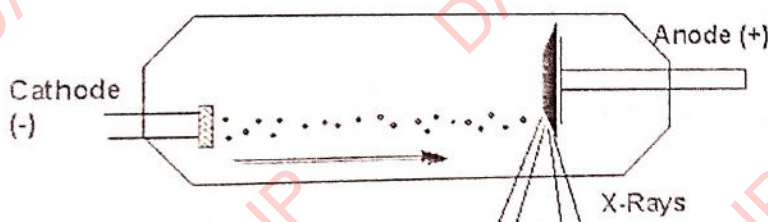
When light emitted from a gas source passes through a prism some discrete sharp lines are obtained on the other side of completely dark back ground. Each line corresponds to a definite wave length such type of spectrum is referred as line or discontinuous spectrum.

"X-RAYS"**ORIGIN OF X-RAYS:**

In 1895, W. C Roentgen found that high-speed cathode rays emitted in a discharged tube when collide with anode, produced a very penetrating radiation. He called it X-rays because it was not known previously. It is also known as Roentgen rays.

MECHANISM OF PRODUCTION:

When the fast moving electrons (cathode rays) collide with the metal anode then due to gain of electrons at anode some amount of energy is released from anode in the form of radiation called X-rays

**PROPERTIES OF X-RAYS:**

1. X-rays are electromagnetic radiations.
2. Their velocity is equal to the velocity of light.
3. Their frequency is very high.
4. The frequency of X-rays depends on the material used at anode.
5. They can affect the photographic plate.
6. They are highly energetic.
7. They can ionize the gas to positive ions by removing electrons.

TYPES OF X-RAYS:

Moseley used different anodes and analyzed the intensity of X-rays. On the basis of wavelengths there are two main types of spectral line of X-rays.

(a) **K-series:** These are produced by the elements (anodes) having large atomic number. These spectral lines have short wavelengths and high energy because transition of electrons occur from high energy level to low energy level (means there is big difference between two energy levels)

(b) **L-series:** Anodes having small atomic number produce these long wavelength spectral lines. These rays have low energy because there is small difference between two energy levels.

"Moseley's Law and Atomic Number"

Moseley in 1913 A.D., comprehensively studied the different wavelengths of x-rays. During his work he noticed that the wavelengths of x-rays decreased regularly with the increase of atomic masses of anode metals. He further observed that the frequencies of these radiations are directly proportional to the number of proton in the nucleus.

STATEMENT:

Moseley law states that *"The square root of the frequency ($\sqrt{u}$) is directly proportional to the atomic number (Z of an element)"*.

$$\sqrt{u} \propto (Z-b)$$

$$\sqrt{u} = a(Z-b)$$

Where a and b are constant

RELATIONSHIP BETWEEN THE WAVELENGTH AND THE NUCLEAR CHARGE OF THE ATOM:

In 1913 Mosley obtained different wavelength when anode made up of different metals were used. According to his observation, the wavelength of X-rays decreases regularly with the increase in positive charge in the atom. Hence each element was assigned a characteristic number equal to the positive charge in the nucleus of its atom. He called it 'Atomic number', and is denoted by ' Z '.

"ATOMIC RADIUS"**DEFINITION:**

"The distance from the centre of the nucleus and the point upto which the outer most electron of the atom is extended"

Difficulties to Measure the Exact Atomic Radius Are;

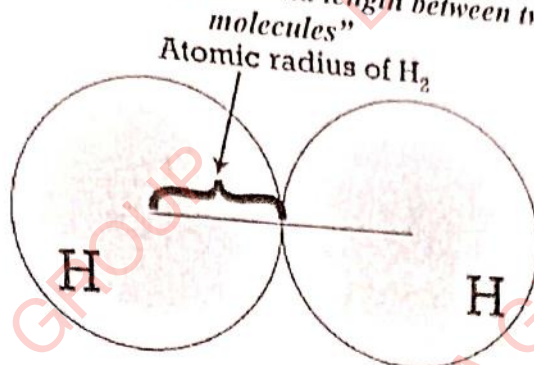
1. An atom is unstable. it cannot be isolated to measure its radius.
2. The exact position of outer most electron is uncertain (Heisenberg's uncertainty principle)

3. The electron density around an atom is affected by the presence of neighboring atom.

Thus, the atomic radius as measured by electron diffraction has only the most probable value.

Radius of Homo diatomic Molecule:

"The radius of an atom is taken as half the bond length between two homonuclear diatomic molecules"



EXAMPLE:

- The distance between two atoms of carbon in diamond is $1.54 \text{ \AA}$
The radius of Carbon $= 1.54/2 = 0.77 \text{ \AA}$
- The intermolecular distance between two Sulphur atoms is $2.060 \text{ \AA}$
The radius of Sulphur $= 2.060/2 = 1.03 \text{ \AA}$

Radius of Hetero diatomic Molecule:

"In case of heterogeneous diatomic molecule like AB, the bond length is $(r_A + r_B)$ and if any one of the radii is known, the other can be calculated"

"IONIC RADIUS"

DEFINITION:

"The distance between the nucleus of an ion and the point upto which nucleus has influence of its electron cloud"

EXPLANATION:

- When a neutral atom loses one or more electrons. Positively charged ion is obtained. The ion is called cation.



- When an atom gains one or more electrons, negatively charged ion is produced. It is called Anion.



Cation Radius:

Cations have smaller radii than neutral atoms because on removal of an electron the effective charge on the nucleus increases and pulls the remaining electrons more firmly.

Radius of Na atom $= 1.57 \text{ \AA}$

Radius of Na^+ ion $= 0.95 \text{ \AA}$

Anion Radius:

Anions have larger radii than the neutral atoms as an excess of negative charge results in greater electron repulsion.

Radius of Cl atom $= 0.99 \text{ \AA}$

Radius of Cl^- ion = $1.81 \text{ \AA}$ **Iso-Electronic ions:**

The iso-electronic ions are the ions having same electronic configuration.

EXAMPLE

Na^+ , Mg^{++} and Al^{+++} have same electronic configuration, i.e., $1s^2, 2s^2, 2p^6$. In these ions the radii decrease with the increase in nuclear charge.

ION	Na^+	Mg^{++}	Al^{+++}
ATOMIC NUMBER	11	12	13
IONIC RADIUS	$0.95 \text{ \AA}$	$0.65 \text{ \AA}$	$0.50 \text{ \AA}$

"PAULI'S EXCLUSIVE PRINCIPLE"**INTRODUCTION:**

It is an empirical rule but agrees fully with experimental observations. It was enunciated by Wolfgang Pauli in 1925.

STATEMENT:

"In an atom no two electrons can have the same set of four quantum numbers".

EXPLANATION:

In an atom, electrons may have maximum number of three quantum number same but the fourth quantum number must be different.

EXAMPLE OF HELIUM:

Consider an example of helium atom. The atomic number of helium is 2. Therefore, it has two electrons in its first orbit. Now,

For the first electron present in $1s$ orbital,

$n = 1, l = 0, m = 0$, but $s = +\frac{1}{2}$ and

For the second electron present in $1s$ orbital,

$n = 1, l = 0, m = 0$, but $s = -\frac{1}{2}$.

Hence it is clear from the above example that two electrons in an orbital must have at least one quantum number different because spin quantum number has two possibilities i.e. $+\frac{1}{2}$ and $-\frac{1}{2}$.

HUND'S RULE OF MAXIMUM MULTIPLICITY:

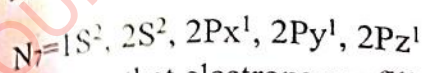
When the degenerate orbitals are available, then

- Electron will enter in different orbitals with parallel spin.
- If orbitals are partially filled then the coming electron will form pair with their electron opposite spin.

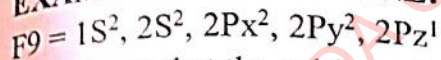
EXPLANATION:

The orbitals present in a sub energy level have equal energies called degenerated orbitals.

EXAMPLE OF NITROGEN:



It shows that electrons are filled in degenerate orbitals with parallel spin.

EXAMPLE OF FLUORINE:

This shows that the pairing of electron start when number of electron is greater than the number of degenerate orbitals.

"OBJECTIVES"**FILL IN THE BLANKS:**

1. The maximum number of electrons in 2p orbital are _____.
2. 3d orbital has _____ energy than 4s orbital.
3. _____ rays are non-material in nature.
4. Charge to mass ratio of cathode rays resembles to that of _____.
5. _____ rays are most penetrating.
6. Neutrons have mass equal to that of _____.
7. Energy is _____ when an electron jumps from higher to lower orbit.
8. Second Ionization Potential has _____ value than the First Ionization Potential.
9. Electronegativity _____ from left to right in a period of Periodic Table.
10. Ionic radii of anions are _____ than the atoms from which they are formed.
11. The velocity of alpha rays is nearly _____ of velocity of light.
12. Natural Radioactivity is confined in _____ elements.
13. Ionic radii of cations are _____ than the atoms from which they are formed.
14. Two electrons with the _____ spin, can never occupy the same atomic orbital.
15. 'Al' has electronic configuration _____.
16. In a group of Periodic Table, the ionization potential _____ from top to bottom as the size of atom increases.
17. Ionization potential values _____ from left to right in a period.
18. The energy required to remove the most loosely bond electron from an atom in gaseous state is called _____.
19. Energy and frequency are _____ proportional to each other
20. The electrons with _____ spin occupy the same orbital
21. The particles of Cathode rays possess _____ charge.
22. The negatively charged particles found in Cathode rays are named as _____.
23. Positive rays are emitted from _____.
24. _____ rays are also known as Canal rays.
25. _____ consists of helium ions and are doubly positively charged.
26. _____ rays consist of negatively charged particles.
27. p-orbitals are _____ shaped
28. The phenomenon in which a stable element is made radioactive by artificial disintegration is called _____.
29. The electron move around the nucleus in different circular paths called _____.
30. The maximum number of electron in a shell is determined by the formula _____.
31. _____ has higher ionization power as compared to β -rays.

32. Neutrons were discovered by _____ in 1932 by the bombardment of beryllium with alpha particles.
33. Number of protons of an element also indicates its _____.
34. According to _____ Principle electrons are fed in the order of increasing orbital energy.
35. According to _____ electrons are distributed among the orbitals of a sub shell to give maximum number of unpaired electron and have same spin.

TRUE AND FALSE:

- Line spectrum is the characteristic of atoms.
- Goldstein in 1886 discovered neutrons by the bombardment of beryllium with α -particles.
- Unlike cathode rays, positive rays do not originate from the anode but are formed due to the ionization of the gas in the discharge tube.
- The maximum number of electrons in any energy level is given by $2n^2$.
- Positive rays like cathode rays have same values of e/m .
- Natural Radioactivity is confined to heavy elements.
- The atom of Oxygen ($Z = 8$) is smaller than the atom of sulphur ($Z = 16$).
- s and p orbitals have same shape.
- Positive rays travel from cathode to anode.
- Ionic radii of cations are smaller than the atoms from which they are formed.
- Line spectrum is obtained from sunlight.
- Oxygen is more electronegative than chlorine.
- First ionization potential has higher value than the second ionization potential value for the element.
- Electronegativity increases from left to right in a period of the Periodic table.
- α -rays are the most penetrating of all the three types of rays.
- Whole mass of an atom is concentrated in the nucleus.
- γ -rays consist of neutral particles.
- Radioactive radiations can affect the photographic plate in dark.
- Orbitals are the regions in which the probability of finding electron is minimum.
- Electron Affinity is the energy required to remove an electron from an atom in the gaseous state.
- Electrons are distributed among the orbitals of a sub-shell to give maximum number of unpaired electrons and have same spin.
- No two electrons in the atom can have all the four quantum numbers.

MULTIPLE CHOICE QUESTIONS:

- The charge on an electron is _____.
(-2.46×10^{-19} coulombs, -1.6×10^{-19} coulombs, 1.6×10^{-19} coulombs)
- The maximum number of electrons that can accommodate by a p-orbital is given by _____.
(2, 6, 10)
- A proton is _____.
(A helium ion, a positively charged particle of mass 1.67×10^{-27} kg, a positively charged particle of mass $1/1837$ that of Hydrogen atom)

4. Most penetrating radiation of a radioactive element is _____.
(α -rays, β -rays, γ -rays)
5. The fundamental particles of an atom are _____.
(Electrons and protons, electrons and neutrons, Electrons, Protons, Neutrons)
6. The mass number of an atom is equal to _____.
(The number of protons, the number of neutrons, the sum of protons and neutrons)
7. "No two electrons in the same atoms can have identical set of four quantum numbers." This statement is known as _____.
(Pauli's Exclusion Principle, Hund's rule, Aufbau Rule)
8. _____ has the highest electronegativity value.
(Fluorine, Chlorine, Bromine)
9. Principle Quantum number describes _____.
(Shape of orbital, size of the orbital, Spin of electron in the orbital)
10. Canal rays are produced from _____.
(Anode, Cathode, Ionization of gas in the discharge tube)
11. Electromagnetic radiation produce from nuclear reactions are known as _____.
(α -rays, β -rays, γ -rays)
12. Cathode rays consist of _____.
(Electrons, Protons, Positrons)
13. The properties of cathode rays _____ upon the nature of the gas inside the tube.
(Depend, partially depend, do not depend)
14. Canal rays consist of _____ particles.
(Negative, Positive, Neutral)
15. Atomic mass of an element is equal to the sum of _____.
(Electrons and protons, protons and neutrons, electrons and neutrons)
16. Neutrons were discovered by _____.
(Faraday, Dalton, Chadwick)
17. The value of Plank's constant is _____.
(6.626×10^{-34} , 6.023×10^{24} , 1.667×10^{-28})
18. P-orbitals are _____ in shape.
(Spherical, diagonal, dumb bell)
19. The removal of an electron from an atom in gaseous state is called _____.

- (Ionization energy, Electron Affinity, Electronegativity)
20. The energy released when an electron is added to an atom in the gaseous state is called _____
(Ionization Potential, electron Affinity, Electronegativity)
21. The power of an atom to attract a shared pair of electrons is called _____
(Ionization Potential, Electron Affinity, Electronegativity)
22. Electronegativity of Fluorine is arbitrarily fixed as _____.
(2, 3, 4)
23. The energy difference between the shells go on _____ when moved away from the nucleus.
(Increasing, decreasing, equalizing)
24. _____ discovered that the nucleus of an atom is positively charged.
(William Crooke's, Rutherford, Dalton)
25. Isotopes are atoms having same _____ but different _____.
(Atomic weight, Atomic number, Avogadro's number)
26. The angular momentum of an electron revolving around the nucleus of atom is _____.
($nh/2\pi$, $n^2h^2/2\pi$, $nh^3/3\pi$)
27. The wavelengths of X-rays are mathematically related to the _____ of anti-cathode element.
(Atomic weight, atomic number, Avogadro's number)
28. Lyman Series of spectral lines appears in the _____ portion of spectrum.
(Ultraviolet, Infra-red, Visible)
29. According to _____ electrons are always filled in order of increasing energy.
(Pauli's Exclusion Principle, Uncertainty Principle, Aufbau Principle)

THEORY QUESTIONS

- Which experiment indicates the existence of electrons and how?
- What are cathode rays? How are their properties proved?
- How are positive rays generated in Cathode-Ray tube? Did they depend on the nature of the gas filled?
- What happens when a current of high voltage is passes through gases at low pressure in the discharge tube? Prove that these rays.
 - Travel in straight line.
 - Are materials in nature?
 - Are electrons.
- Write a short note on Radio activity.
- Describe the characteristic of Alpha, Beta and Gamma rays.
- How were neutron discovered? (Give the reaction)
- Give postulates of Rutherford's atomic model.
- What are the weaknesses of Rutherford's atomic theory?
- Write the postulates of Bohr's Atomic Theory. Derive an expression for the energy of electron in an atom.
- Derive an expression for radius of hydrogen atom?
- Write down the name of the spectral lines of a hydrogen atom.
- Differentiate between Lyman and Balmer series.
- Explain Planks quantum theory in details.
- Differentiate between line and continuous spectrum.
- How x-rays are produced. How were they discovered?
- Discuss the four quantum numbers in detail. Explain location and properties of electron.
- Write down the values of all the four quantum numbers for each of the electron of Helium atom.
- Differentiate between Orbit and orbital.
- Draw the shape of the s and p- orbital.
- State Pauli's Exclusion principle; explain it by giving Helium atom.
- State and explain Hund's Rule.
- Give reasons for the following :
 - Ionization potential of oxygen is smaller than that of nitrogen
 - An orbital have two electrons with opposite spin
 - Why does electron affinity of chlorine is greater than fluorine?
 - Differentiate between Electron negativity and Electron affinity.

NUMERICALS FROM PAST PAPERS

BOHR'S MODEL:

1. Derive the expression for the frequency and wave number of radiation when electron jumps from higher orbit n_2 to lower orbit n_1 . The energy of electron is $E = \frac{-2\pi^2 m Z^2 e^4}{n^2 h^2}$
2. Calculate the wave number of line in Lyman series when an electron jumps from 3rd orbit to the 1st orbit. ($R_H = 109678 \text{ cm}^{-1}$) **97503.74 cm⁻¹**
3. Bohr's radius is $0.529 \text{ \AA}$. Find the radius of 3rd orbit of hydrogen atom. **4.761⁰ \AA**
4. Calculate the wave number of spectral lines of hydrogen gas when the electron jumps from $n=4$ to $n=2$. ($R_H = 109678 \text{ m}^{-1}$) **20564.62 cm⁻¹**
5. Derive the expression for the radius of n th orbit of hydrogen atom using Bohr's postulate.
6. Calculate the wave number of an electron when it jumps from an orbit $n=5$ to an orbit $n=1$. **105290.0 cm⁻¹**
7. Derive the expression for energy of an electron in n th orbit.
8. Bohr's radius is $0.529 \text{ \AA}$. Find the radius of 2nd orbit of hydrogen atom. **R=2.116⁰ \AA**
9. Calculate the angular momentum of electron in first energy level of hydrogen atom. ($h = 6.625 \times 10^{-27} \text{ erg. Sec}$) **1.054 * 10⁻²⁷ erg sec**
10. Calculate the wave number of radiation emitted in Lyman series of hydrogen atom when electron jumps from 2nd to 1st orbit. ($R_H = 109678 \text{ cm}^{-1}$)
11. Calculate the radius of 3rd orbit of hydrogen atom using Bohr's radius as $a_0 = 0.529 \text{ \AA}$.
12. Calculate the wave number of an electron in hydrogen atom when it jumps from an orbit $n=3$ to an orbit $n=2$.
13. Calculate the radius of orbit $n=3$ for a hydrogen atom in Angstrom unit. ($h = 6.625 \times 10^{-27} \text{ erg. sec}$, $m = 9.11 \times 10^{-28} \text{ gm}$, $e = 4.8 \times 10^{-10} \text{ esu}$).
14. Calculate the wave number of line in Lyman series when an electron jumps from orbit $n=3$ to orbit $n=1$.
15. Starting from $\Delta E = E_2 - E_1$ derive the expression for wave number for the hydrogen atom.

ELECTRONIC CONFIGURATION:

1. Write the electronic configuration of the following:
 - a. Mo ($Z=42$)
 - b. Ag ($Z=47$)
2. Arrange the orbitals in the order of ascending energy according to $(n+l)$ rule: 3d, 4s, 5s, 6s.
3. Write the electronic configuration of the following and also give the number of electron and proton in each:
 - a. Cu ($Z=29$)
 - b. Mg^{+2} ($Z=12$)
 - c. Cl^{-1} ($Z=17$)
4. What rules and principles are violated in the following electronic configuration:

*1s², 2s³ *1s², 2p² *1s², 2s², 2p⁶, 3s², 3p⁶, 3d⁴
5. Write the correct sequence of the orbitals for the electronic configuration according to $(n+l)$ rule:

3d, 4s, 4p, 4d, 5s, 5p

6. Write the ground state and excited state configuration of carbon atom.
7. Write the electronic configuration of Na ($Z=11$) and B ($Z=5$). What are the values of n and l for the last sub-shell of each element?
8. Arrange the following energy levels in of ascending order according to $(n+l)$ rule:
 $4d, 7s, 4f$
9. Write the electronic configuration of S^{-2} .
10. Which principle is violated in the following electronic configuration:
 $1s^2, 2s^1, 2p^5$ $1s^2, 2s^3, 2p^5$
11. The electronic configuration of 7N : $1s^2, 2s^2, 2p_x^2, 2p_y^1, 2p_z^0$, which rule is violated?
12. Write the electronic configuration of: ${}_{13}Al^{+3}$, ${}_{29}Cu$, ${}_{17}Cl^{-1}$
13. Which rule or principle is violated in the following electronic configuration:
 $1s^{11}, 2p_x^{11}, 2p_y^1, 2p_z^0$
14. Write the electronic configuration of: ${}_{29}Cu$, ${}_{19}K^+$, ${}_{8}O^{-2}$
15. Write the electronic configuration for the ground state of the following ; $Z=19$, $Z=21$, $Z=16$
16. Write down the electronic configuration of: Cr ($Z=24$), Li^+ ($Z=3$), and F^- ($Z=9$).
17. Make the correct sequence of the orbitals for the electronic configuration according to $n+l$ rule.
 $4p, 4s, 3d, 5s, 5p$
18. The electronic configuration of C ($z=6$) is $1s^2, 2s^3, 2p^1$. Make correction with reference to the respective law or principle.
19. Write the electronic configuration of the following:
 Cr ($z=24$); Cl^- ($z=17$)
20. What rules or principals are violated in the following electronic configuration:
a. $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4d^4, 4s$ b. $1s^2, 2s^2, 2p^6, 3s^2, 3p_x^{11}, 3p_y^0, 3p_z^0$
21. Make the correct sequence of the orbitlas for the electronic configuration on the basis of $n+l$ rule:
 $3d, 4s, 4p, 4d, 5s, 6s, 5p$
22. Write the electronic configuration of the following:
* Cr ($z=24$) * Mo ($z=42$)

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